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**INTRODUCTION TO BUSINESS ANALYTICS BANA (607)**

**Prof. Pouyan Eslami**

**THE GROWING DEMAND FOR CANNABIS - A TIME SERIES CASE STUDY**

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## INTRODUCTION:

Cannabis is also called marijuana; it is a plant used throughout history and in various ways, whether for medical, recreational, or industrial purposes. The plant contains chemical compounds called cannabinoids such as THC (tetrahydrocannabinol) and CBD (cannabidiol). These two compounds are used for its psychoactivity and therapeutical effects of the drug. Cannabis is a commonly consumable product, ranging from dried flowers to oils, edibles, and concentrates.

The legality of cannabis within a country will vary across the world and will be dependent on cultural, medical, and legal grounds. For instance, many countries within the past century have considered cannabis illegal because of psychoactive concerns. However, the shift is now global, with most nations and U.S. states decriminalizing or legalizing cannabis either for medical or recreational use. Today, cannabis is legal in selective regions under strict regulations, including age restrictions, quality control norms, and regulated sale. For instance, in the United States, cannabis is legal under the medical use in many states and for recreation in some states, even though it remains illegal at federal levels.

The dataset provides a detailed transaction data within different dispensaries in the United States. It captures essential records of the customer profiles, geographical locations, the products offered, and orders, thus creating a picture of consumer behaviour and purchase patterns within the cannabis marketplace.

Data includes transactions of cannabis from 2019 to 2023 covering a variety of cannabis goods, such as flower, extracts, topicals, and vape cartridges. It simultaneously spans all states with standard categories of goods at various price levels while tracking customer preferences and spending habits. Hence, the data show where Cannabis's customers are from all over the country-from orders shipped within Oregon, Arizona, California, and Nevada. The report intends to analyse data on customer orders and extract trends from it, such as patterns in popularity of products, differences between geographical areas and customer demographics, as well as others intended to provide useful insights that could be applied by Cannabis and such businesses to enhance the offer of products, reach the right consumers and optimize their operational strategy within the rapidly growing cannabis market.

## RESEARCH QUESTION:

How do customer orders vary over time, and can we forecast future sales based on past trends?

## STEPS:

- **Data Preparation and Analysis**

We performed data cleaning to make the dataset more and suitable for predictive modeling, enhancing the reliability of any insights derived from the analysis. The following code analyses and cleans a dataset containing cannabis order data for time series analysis. The purpose is to identify and remove outliers to ensure the data's integrity before modelling. We have also grouped the dataset by ord\_datetime to calculate daily order counts.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
import statsmodels.api as sm
import numpy as np

# Load your dataset
df = pd.read_csv('customerOrders.csv')

# Convert ord_datetime to datetime format
df['ord_datetime'] = pd.to_datetime(df['ord_datetime'])

# Group by day and count the number of orders
df['ord_date'] = df['ord_datetime'].dt.date # Extract the date part
orders_per_day = df.groupby('ord_date').size().reset_index(name='order_count')

# Merge the orders_per_day back into the original DataFrame
df = df.merge(orders_per_day, on='ord_date', how='left')

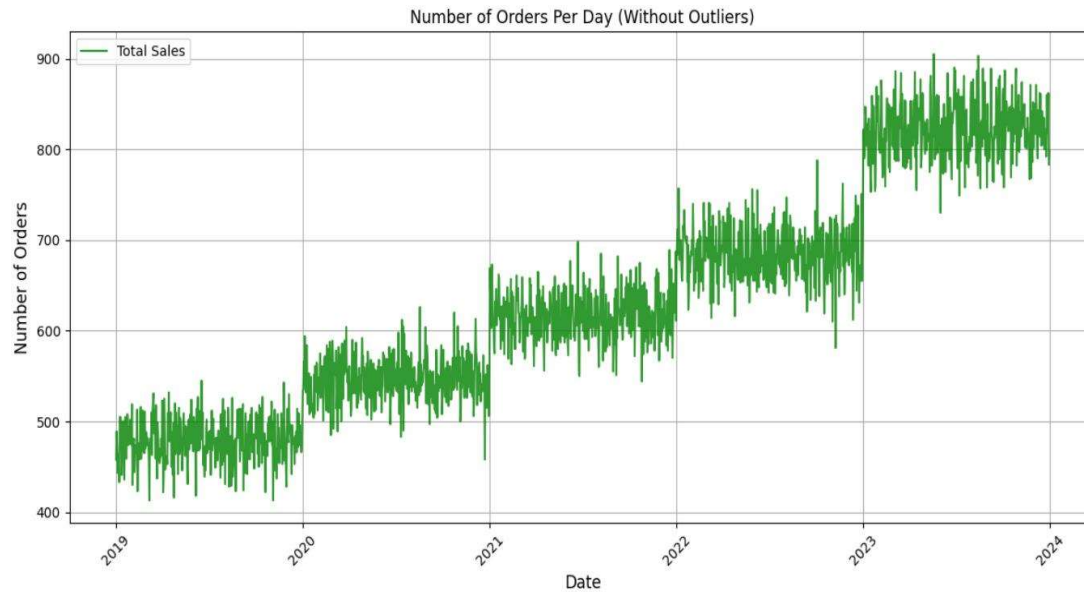
# Calculate Q1 (25th percentile) and Q3 (75th percentile)
Q1 = orders_per_day['order_count'].quantile(0.25)
Q3 = orders_per_day['order_count'].quantile(0.75)
IQR = Q3 - Q1

# Define outlier bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Remove outliers
orders_per_day_no_outliers = orders_per_day[(orders_per_day['order_count'] >= lower_bound) & (orders_per_day['order_count'] <= upper_bound)]
```

- **Exploratory Analysis:**

For better understanding of trends in cannabis order volumes over time we have visualized the clean dataset. A line chart is used here to represent the number of daily cannabis orders (y-axis) over time (x-axis) as a continuous trend. Each data point (representing a specific day's order count) is connected by a line, highlighting the overall pattern and trends in the data while smoothing out individual fluctuations. This chart effectively conveys changes and growth in cannabis orders over the observed period.



The graph shows a steady upward trend in cannabis orders, reflecting increasing demand over time

- **Modeling:**

A linear regression model is used to identify the trends and predict the future sales.

### Linear Regression Model

Linear regression is simple and effective for identifying and quantifying trends over time. It models the relationship between the dependent variable (number of orders) and the independent variable (time) as a straight line. It requires less computational power compared to more complex models, making it suitable for quick analyses, especially when the data exhibits a clear linear trend. A linear model can be extended to make predictions about future trends, which is valuable for planning inventory, staffing, or logistics.

### Code:

```
from sklearn.linear_model import LinearRegression
import statsmodels.api as sm
import numpy as np

# Prepare data for regression
orders_per_day_no_outliers['ordinal_date'] =
pd.to_datetime(orders_per_day_no_outliers['ord_date']).map(pd.Timestamp.toordinal)
X = orders_per_day_no_outliers[['ordinal_date']] # Independent variable (date as ordinal)
y = orders_per_day_no_outliers['order_count'] # Dependent variable (order count)

# Fit model for Simple Linear Regression
model = LinearRegression()
model.fit(X, y)
print("Coefficient ( $\beta_1$ ):", model.coef_[0])
print("Intercept ( $\beta_0$ ):", model.intercept_)
```

```

# Statistical Significance
X_with_const = sm.add_constant(X)
model_sm = sm.OLS(y, X_with_const).fit()
print(model_sm.summary())

# Plot the regression line
plt.figure(figsize=(12, 6))

plt.scatter(
    orders_per_day_no_outliers['ord_date'],
    y,
    label='Actual Orders',
    color='blue',
    alpha=0.7,          # Make points slightly transparent
    edgecolors='black', # Add a black edge to each point
    linewidths=0.5,     # Make edge lines thin
    s=50               # Adjust marker size
)

# Generate predictions using the fitted model
predictions = model.predict(X)

plt.plot(
    orders_per_day_no_outliers['ord_date'],
    predictions,
    label='Regression Line',
    color='red',
)

plt.xlabel('Date')
plt.ylabel('Number of Orders')
plt.title('Linear Regression: Number of Orders Per Day')
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig("linear_regression_chart.png")
print("Chart saved as 'linear_regression_chart.png'")

```

## Output

Coefficient ( $\beta_1$ ): 0.21672408212806935

Intercept ( $\beta_0$ ): -159306.62080340454

### OLS Regression Results

|                   |                  |                     |           |
|-------------------|------------------|---------------------|-----------|
| Dep. Variable:    | order_count      | R-squared:          | 0.887     |
| Model:            | OLS              | Adj. R-squared:     | 0.887     |
| Method:           | Least Squares    | F-statistic:        | 1.437e+04 |
| Date:             | Thu, 12 Dec 2024 | Prob (F-statistic): | 0.00      |
| Time:             | 12:05:03         | Log-Likelihood:     | -9358.8   |
| No. Observations: | 1826             | AIC:                | 1.872e+04 |
| Df Residuals:     | 1824             | BIC:                | 1.873e+04 |
| Df Model:         | 1                |                     |           |
| Covariance Type:  | nonrobust        |                     |           |

|              | coef       | std err  | t        | P> t  | [0.025    | 0.975]    |
|--------------|------------|----------|----------|-------|-----------|-----------|
| const        | -1.593e+05 | 1334.335 | -119.390 | 0.000 | -1.62e+05 | -1.57e+05 |
| ordinal_date | 0.2167     | 0.002    | 119.862  | 0.000 | 0.213     | 0.220     |

|                |       |                   |          |
|----------------|-------|-------------------|----------|
| Omnibus:       | 2.057 | Durbin-Watson:    | 0.966    |
| Prob(Omnibus): | 0.357 | Jarque-Bera (JB): | 2.060    |
| Skew:          | 0.082 | Prob(JB):         | 0.357    |
| Kurtosis:      | 2.987 | Cond. No.         | 1.03e+09 |

#### Notes:

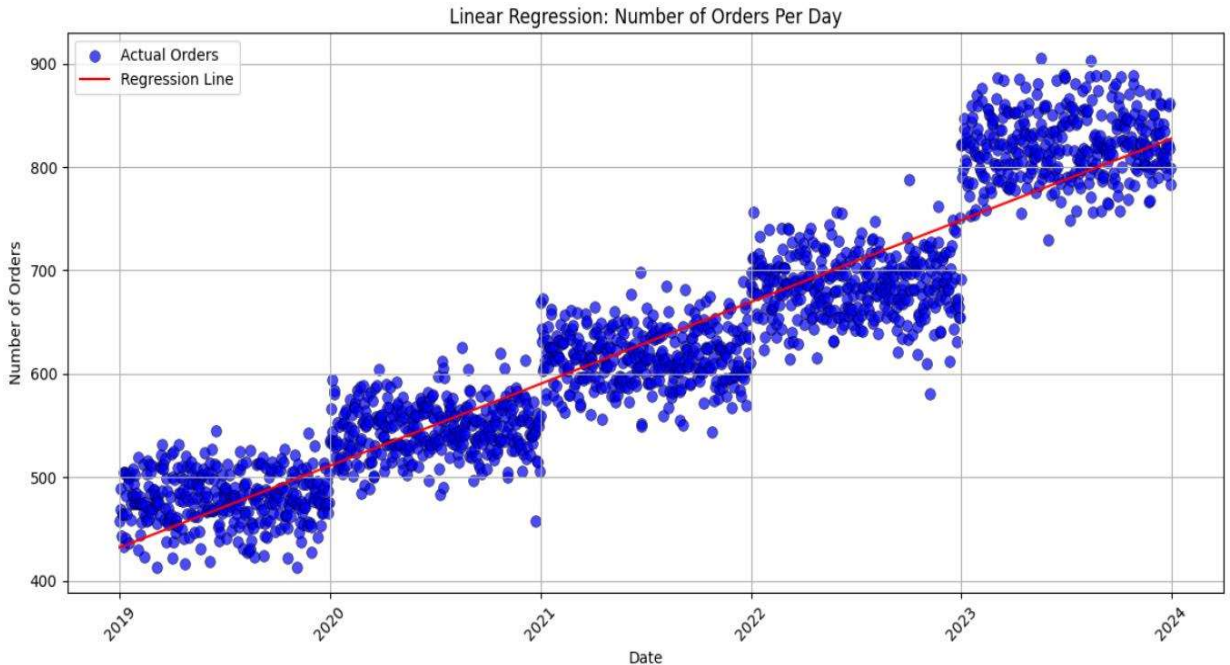
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.03e+09. This might indicate that there are strong multicollinearity or other numerical problems.

Chart saved as 'linear regression chart.png'

The regression output and accompanying graph demonstrate the relationship between daily cannabis orders and time, modeled using Ordinary Least Squares (OLS) regression. The regression results indicate an  $R^2$  value of 0.887, meaning that 88.7% of the variance in daily order counts is explained by the passage of time. The slope coefficient ( $\beta_1=0.2167$ ) shows a significant positive relationship, suggesting that the number of orders increases by approximately 0.22 orders per unit of time. The intercept ( $\beta_0=-159306.62$ ) serves as the baseline when the time variable is zero but lacks practical interpretability in this context. The p-value for the slope is 0.000, confirming the relationship's statistical significance.





This graph represents the results of a linear regression analysis performed on the dataset, which explores the relationship between time (x-axis: Date) and the number of cannabis orders per day (y-axis: Number of Orders). Each blue dot represents the actual number of orders recorded on a particular day. The red line represents the best-fit linear regression model. This line predicts the number of orders based on the date, illustrating the general trend over time. The regression line shows an upward slope, indicating that the number of cannabis orders has been steadily increasing over time.

## FORECASTING AND INSIGHTS

The graph confirms consistent growth in cannabis orders over time, suggesting rising consumer demand. The upward trajectory may reflect increased market penetration, changing regulations, or broader acceptance of cannabis products.

## CONCLUSION

The analysis of daily cannabis orders over time reveals a clear and statistically significant upward trend in demand, with the linear regression model explaining 88.7% of the variability in orders. This strong relationship between time and order count suggests that cannabis consumption has steadily increased during the observed period, likely driven by factors such as market growth, legalization, and changing consumer behaviors. The high R<sup>2</sup> value and significant slope coefficient, makes it a valuable tool for forecasting future demand. These insights can guide dispensaries and suppliers in optimizing inventory, staffing, and strategic planning.